

Supplementary Materials for: MOSAIC: A Dual-Layer Fault-Tolerant Molecular Storage Codec with Empirical Coverage Threshold Characterisation for Nanopore Sequencing

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S1. Half MOSAIC Baseline: Analogue Degradation

The unhardened *Half MOSAIC* baseline demonstrates the exact failure modes of single-layer continuous-stream architectures. As shown in Table S1 and Figure S1, severe frame-shifting (indel) errors bypass the heuristic consensus threshold, creating catastrophic byte-stream misalignment. This produces a visually recognisable image (high SSIM) that completely fails cryptographic verification.

Table S1: Half MOSAIC Baseline Survival and Perceptual Metrics (Ablation Study)

(a) Legacy R9.4.1 Channel					(b) Next-Gen R10.4 Channel				
Cov.	Survival (%)	PSNR (dB)	SSIM	SHA-256	Cov.	Survival (%)	PSNR (dB)	SSIM	SHA-256
1×	0.04	7.96	0.008	False	1×	22.30	9.06	0.019	False
2×	0.19	7.95	0.009	False	2×	34.73	9.81	0.028	False
3×	9.52	8.43	0.013	False	3×	90.82	18.38	0.486	False
4×	38.50	10.07	0.032	False	4×	98.25	25.05	0.863	False
5×	67.34	12.66	0.092	False	5×	99.65	31.33	0.965	False
6×	66.27	12.77	0.093	False	6×	99.90	36.78	0.989	False
7×	85.70	16.50	0.332	False	7×	99.98	47.56	0.999	False
8×	79.80	14.85	0.210	False	8×	99.99	46.41	0.998	False
9×	91.00	18.62	0.506	False	9×	100.00	∞	1.000	False
10×	95.82	21.92	0.725	False	10×	100.00	∞	1.000	False
11×	94.07	20.06	0.620	False					
12×	97.03	23.42	0.799	False					
13×	95.87	21.65	0.707	False					
14×	97.96	24.92	0.855	False					
15×	98.97	27.86	0.923	False					
16×	98.68	25.86	0.884	False					
17×	99.22	28.94	0.940	False					
18×	98.84	27.49	0.915	False					
19×	99.28	30.38	0.955	False					
20×	99.61	33.42	0.977	False					

(a) Legacy R9.4.1 Extreme-Noise Channel



(b) Next-Generation R10.4 Moderate-Noise Channel

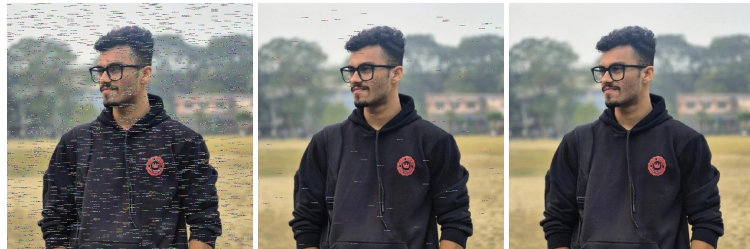


Figure S1: **Half MOSAIC Visual Progression.** (a) Progression under legacy R9.4.1 conditions across 5×, 9×, 12×, 15×, and 20× coverages. Uncaught indels bypass the consensus threshold, inducing severe RGB phase-shifting that resolves gracefully to near-lossless perceptual quality (SSIM = 0.965 at 20×). (b) Progression under R10.4 duplex chemistry across 3×, 5×, and 10× coverages. The superior baseline accuracy achieves an SSIM of 1.000 at 10×. Crucially, SHA-256 verification fails universally across every tier, isolating the exact failure mode of baseline architectures: heuristic metrics mask silent, catastrophic byte-stream corruption.

S2. Full MOSAIC: Deterministic Recovery

Table S2 and Figure S2 demonstrate the deterministic zero-stating behavior of the *Full MOSAIC* dual-layer matrix. Blocks exceeding the 32-frame erasure budget are strictly quarantined to prevent silent corruption, rendering as black voids until the critical coverage threshold is breached, yielding a mathematically guaranteed, bit-perfect SHA-256 match.

Table S2: Full MOSAIC Dual-Layer Architecture: Deterministic Erasure Clearance and Cryptographic Convergence

Channel	Cov.	Survival (%)	Erasures	PSNR (dB)	SSIM	SHA-256
R9.4.1	5×	64.86	6,631	4.08	0.000	FAIL
R9.4.1	7×	83.18	3,173	4.21	0.010	FAIL
R9.4.1	8×	76.32	4,469	4.08	0.000	FAIL
R9.4.1	9×	89.53	1,976	13.21	0.800	FAIL
R9.4.1	10×	94.32	1,071	∞	1.000	VERIFIED
R9.4.1	12×	96.43	673	∞	1.000	VERIFIED
R10.4	3×	90.45	1,803	17.11	0.912	FAIL
R10.4	4×	98.25	330	∞	1.000	VERIFIED
R10.4	5×	99.51	92	∞	1.000	VERIFIED

Unlike the *Half* state, *Full MOSAIC* enforces strict block-level zero-stating when the $T = 32$ erasure budget is exceeded. Convergence to absolute file integrity is achieved strictly at 10×

 (R9.4.1 extreme-noise) and 4× (R10.4 moderate-noise) coverage.

(a) Deterministic Integrity Under Legacy R9.4.1 Channel



(b) The Enterprise Coverage Floor Under R10.4 Channel



Figure S2: **Full MOSAIC Visual Progression and Cryptographic Convergence.** (a) At low coverages (5×

 under the R9.4.1 channel, blocks exceeding the $T = 32$ erasure budget safely zero-state, rendering as black voids (SSIM = 0.000) rather than corrupting the byte-stream. At the critical 10× threshold, the outer matrix clears all remaining erasures, converging to SHA-256-verified, bit-exact recovery. (b) By strictly 4× coverage under R10.4, erasure totals drop safely below the block budget, achieving a mathematically guaranteed, bit-perfect file.

S3. Extended Statistical Validation of the Consensus Anomaly

In the main text (Section V-D), we identify a non-monotonic "sawtooth" degradation in inner-layer survival within the $10\times$ – $13\times$ coverage zone under R9.4.1 channel conditions. To rigorously confirm that this instability is a fundamental algorithmic property of the fractional consensus ceiling function ($V_{\text{req}} = \lceil 0.4 \times C \rceil$) rather than an isolated artefact of the primary payload, an extended multi-trial statistical sweep ($n = 5$ trials per point) was conducted.

As detailed in Table S3, the empirical variance across five independent stochastic channel simulations remains strictly bounded ($\sigma \leq 0.24\%$). The persistence of the survival drops at $11\times$ and $13\times$ across all trials confirms the structural reality of the mathematical strictness penalty. Critically, the 100% SHA-256 verification rate demonstrates the outer matrix's capacity to absorb extreme algorithmic variance without file corruption.

Table S3: Empirical Variance of Inner Survival (p) Across the Non-Monotonic Zone

Coverage	Mean Survival ($\bar{p}$)	Std. Dev. (σ)	SHA-256 Verified
$10\times$	94.50%	0.12%	5/5 trials
$11\times$	92.16%	0.24%	5/5 trials
$12\times$	96.20%	0.16%	5/5 trials
$13\times$	94.12%	0.07%	5/5 trials
$14\times$	97.13%	0.17%	5/5 trials

Calculated over $n = 5$ independent simulation trials using a secondary validation payload. The continued oscillation in mean survival perfectly mirrors the theoretical strictness penalties derived in the main text.

S4. Stochastic Markov Channel Model Parameters

To ensure the reproducibility of the empirical coverage thresholds (C^*), the Python-based simulation environment models the complete biological data lifecycle using a first-order Markov process. The simulator is intentionally calibrated to upper-bound the empirical error rates reported in literature for Oxford Nanopore flow cells, ensuring the MOSAICarchitecture is stress-tested against worst-case physical conditions.

1. Legacy R9.4.1 Extreme-Noise Channel Profile:

Raw reads on legacy flow cells exhibit total error rates between 6% and 12%. The simulator establishes a severe, indel-heavy profile:

- **Substitution Probability (p_s):** 3.0%
- **Insertion Probability (p_i):** 1.0%
- **Deletion Probability (p_d):** 2.0%

2. Next-Generation R10.4 Moderate-Noise Channel Profile:

The R10.4 chemistry features a longer nanopore barrel and improved motor protein control, reducing total raw error to $\sim 1 - 5\%$. The simulator establishes the following upper-bound simplex profile:

- **Substitution Probability (p_s):** 0.5%
- **Insertion Probability (p_i):** 0.2%
- **Deletion Probability (p_d):** 0.4%

3. The Homopolymer Deletion Penalty:

To accurately reflect the physical physics of the read-head, the simulator implements a dynamic penalty for low-complexity regions. Any pathological data run consisting of identical sequential bases of length $L \geq 3$ (e.g., AAA, CCC) incurs a strict $2\times$ multiplier to its baseline deletion probability ($p_d \times 2$). This simulates the well-documented tendency of the nanopore motor protein to "slip" or accelerate through repetitive sequences without registering discrete ionic current shifts. The 100% compliance rate of the MOSAICLFSR data-whitening layer (which restricts $L \leq 3$) successfully mitigates this physical vulnerability before synthesis.